Write a paper on the principle that nano-TiO2 enhances the mechanical properties of NBR rubber

Nano-TiO_2 as a Reinforcing Filler for NBR Rubber: Mechanisms of Mechanical Enhancement

Abstract

Nitrile butadiene rubber (NBR) is a polar elastomer widely used in industrial applications due to its excellent oil resistance, but it suffers from inherently low mechanical strength due to the absence of strain-induced crystallization. The incorporation of nano-titanium dioxide (TiO_2) has emerged as an effective strategy for enhancing the mechanical performance of NBR composites. This paper reviews the fundamental principles governing the reinforcing effects of nano-TiO_2 in NBR, including filler dispersion, interfacial interactions, crosslinking behavior, and the influence of filler loading on mechanical properties. Surface modification strategies and the critical roles of filler-rubber interactions are discussed in detail to provide a comprehensive understanding of the reinforcement mechanisms.

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1. Introduction

Acrylonitrile butadiene rubber (NBR) is a copolymer of acrylonitrile and butadiene that exhibits beneficial properties including a broad temperature application range, excellent oil and solvent resistance, good bonding performance, and relatively low cost . However, NBR does not exhibit self-reinforcing behaviors due to its irregular random copolymer structure, which prevents strain-induced crystallization and results in comparatively weak mechanical properties .

The incorporation of reinforcing fillers is essential to achieve high-performance NBR products. Among various fillers, titanium dioxide (TiO_2) nanoparticles have attracted considerable attention due to their unique characteristics, including non-toxicity, high reactivity, chemical stability, biocompatibility, and low cost . TiO_2 has been widely employed as a white pigment in plastics and rubber, but recent research has demonstrated its potential as an effective reinforcing filler that can significantly improve the mechanical properties of NBR composites .

This paper examines the fundamental principles by which nano-TiO_2 enhances the mechanical properties of NBR rubber, focusing on dispersion characteristics, interfacial interactions, crosslinking effects, and the critical role of filler loading.

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2. Fundamental Mechanisms of Reinforcement

2.1 The Challenge of NBR Reinforcement

NBR presents unique challenges for reinforcement due to its molecular structure. As a random copolymer of acrylonitrile and butadiene, NBR lacks the regular structure necessary for strain-induced crystallization, which is the primary reinforcement mechanism in natural rubber . Consequently, the mechanical properties of unfilled NBR are substantially inferior to those of self-reinforcing elastomers, necessitating the use of reinforcing fillers to achieve acceptable performance for engineering applications.

2.2 Dispersion of Nano-TiO_2 in NBR Matrix

The dispersion of nanoparticles within the rubber matrix is the most critical factor governing the reinforcing effect. Nano-TiO_2 particles, due to their small size (typically 20–40 nm) and high surface energy, tend to agglomerate unless properly dispersed. Research has shown that when nano-TiO_2 is well-dispersed in NBR, with aggregate sizes predominantly less than 100 nm, significant mechanical property improvements can be achieved .

The quality of dispersion directly influences:

- The effective surface area available for interaction with the rubber matrix

- The distribution of stress throughout the composite

- The formation of a filler network that contributes to reinforcement

2.3 Interfacial Interactions

The polar nature of NBR, imparted by the acrylonitrile content, creates favorable conditions for interaction with the surface of TiO_2 nanoparticles. The surface of TiO_2 particles contains hydroxyl groups (-OH), including both terminal and bridge hydroxyl groups, which can form physical interactions with the polar -C≡N groups of NBR . This interaction is schematically represented in the literature as a key mechanism for reinforcement .

The filler-rubber interaction contributes to reinforcement through:

Physical Adsorption: The polar groups on NBR chains are attracted to the polar surface of TiO_2, resulting in the formation of bound rubber layers around the filler particles. These immobilized polymer layers effectively increase the volume fraction of the reinforcing phase.

Improved Stress Transfer: Strong interfacial interactions enable efficient transfer of stress from the rubber matrix to the rigid TiO_2 particles, enhancing the overall mechanical performance of the composite.

2.4 Surface Modification Strategies

Surface modification of nano-TiO_2 has been developed to further enhance dispersion and interfacial interactions in NBR composites. A combined covalent and noncovalent modification approach has demonstrated significant promise . In this method:

1. **Noncovalent Modification**: TiO_2 nanoparticles are first coated with poly(catechol/polyamine) (PCPA) through noncovalent deposition. This approach preserves the original surface characteristics of the nanoparticles while providing a platform for subsequent functionalization .

2. **Covalent Modification**: The PCPA-coated nanoparticles are then functionalized with γ-(2,3-epoxypropoxy)-propytrimethoxysilane (GPTMS), which introduces epoxy groups that provide strong interfacial interactions with the polar NBR matrix .

This dual-modification strategy has been shown to enhance mechanical, dielectric, and electromechanical properties of NBR composites. At a loading of 10 phr modified TiO_2, the composite exhibited actuated strain approximately 230% of pure NBR .

2.5 Effects on Crosslinking and Cure Characteristics

The addition of nano-TiO_2 significantly influences the vulcanization behavior of NBR, which in turn affects mechanical properties. Research has demonstrated that TiO_2 functions as a co-activator in sulfur curing systems, stimulating crosslinking reactions .

Key observations regarding cure characteristics include:

Parameter	Effect of TiO_2 Addition	Interpretation
Scorch time (ts1)	Decreased	Faster initiation of crosslinking
Optimum cure time (tc90)	Decreased	Accelerated overall curing process
Cure rate index (CRI)	Increased up to optimum	Enhanced cure kinetics
Minimum torque (ML)	Increased	Higher compound viscosity, restricted molecular mobility
Maximum torque (MH)	Increased	Higher crosslink density and composite stiffness
Torque difference (MH-ML)	Increased	Indicates higher crosslink density

The increase in torque difference (MH−ML) is particularly significant as it indirectly reflects the crosslink density in rubber vulcanizates. This increase indicates that TiO_2 contributes to a higher crosslink density through strong filler–rubber interactions . The physical interactions between polar -C≡N groups in NBR

and -OH groups on TiO_2 surfaces are proposed to create additional physical crosslinks that enhance the overall crosslink density.

2.6 Role of Filler Loading

The concentration of nano-TiO_2 is a critical parameter that determines the reinforcing effect. Studies investigating TiO_2 content ranging from 0 to 110 phr have revealed important trends .

Optimal Loading: The optimal TiO_2 content for maximum tensile strength and elongation at break was found to be 90 phr . At this loading, the tensile strength of NBR/TiO_2 composites was increased by 144.8% compared to pure NBR, while elongation at break increased by 40.1% .

Reinforcement Mechanism: The significant improvement in mechanical properties at optimal loading is attributed to:

- Good dispersion of TiO_2 in the NBR matrix, confirmed by scanning electron microscopy

- Strong filler–rubber interactions

- Increased crosslink density

- Effective stress transfer from the rubber matrix to the rigid filler particles

Beyond Optimal Loading: At very high TiO_2 content (110 phr), some properties begin to decline. The minimum torque showed a decreasing trend at 110 phr, attributed to filler agglomeration, which disrupts the reinforcing network . This demonstrates the importance of maintaining optimal filler loading to maximize reinforcement.

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3. Dynamic Mechanical Properties

The incorporation of nano-TiO_2 also affects the dynamic mechanical properties of NBR composites, providing further evidence of reinforcement.

3.1 Glass Transition Temperature (Tg)

Dynamic mechanical thermal analysis (DMTA) has revealed that the addition of TiO_2 shifts the glass transition temperature of NBR to higher temperatures . This shift indicates restricted segmental motion of the polymer chains due to:

- Physical interactions between polymer chains and filler surfaces

- Increased crosslink density

- Formation of bound rubber layers

3.2 Storage Modulus

TiO_2-filled NBR composites exhibit higher storage modulus compared to unfilled NBR . The increase in storage modulus is attributed to:

- The high intrinsic stiffness of TiO_2 particles

- The formation of a filler network that provides additional load-bearing capacity

- Strong interfacial interactions that enhance the effective stiffness of the polymer matrix

3.3 Damping Behavior

The incorporation of TiO_2 reduces damping in NBR composites, as indicated by a decrease in tan δ . This reduction is consistent with the formation of a more constrained polymer network with reduced energy dissipation during cyclic deformation.

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4. Mechanisms of Mechanical Enhancement

Based on the available evidence, the enhancement of mechanical properties in nano-TiO_2/NBR composites can be attributed to several interrelated mechanisms:

4.1 Hydrodynamic Reinforcement

The rigid TiO_2 nanoparticles act as dispersed hard inclusions within the rubber matrix. According to hydrodynamic reinforcement theory, the presence of these particles increases the effective volume fraction of the reinforcing phase, contributing to increased modulus and strength.

4.2 Crosslink Density Enhancement

The polar interactions between TiO_2 surfaces and NBR chains create additional physical crosslinks that complement the chemical crosslinks formed during vulcanization. The increased crosslink density, evidenced by higher torque differences, contributes to:

- Higher tensile strength
- Improved modulus
- Reduced creep and set

4.3 Bound Rubber Formation

The strong filler-rubber interactions promote the formation of bound rubber—polymer chains that are immobilized on the filler surface. This bound rubber layer effectively increases the volume fraction of the reinforcing phase and creates a gradient in mechanical properties from the filler surface to the bulk matrix.

4.4 Crack Arrest and Energy Dissipation

The dispersion of rigid TiO_2 nanoparticles provides obstacles to crack propagation. When cracks encounter nanoparticles, they may be deflected, bifurcated, or terminated, requiring additional energy for propagation. This mechanism contributes to improved toughness and resistance to failure.

4.5 Synergistic Effects

Recent research has demonstrated that TiO_2 nanoparticles can be combined with other reinforcing components, such as helical carbon nanofibers, to create hybrid fillers that provide synergistic reinforcement . The anchoring of TiO_2 nanoparticles on the surface of carbon nanofibers through covalent bonding has been shown to:

- Improve the dispersion of TiO_2 nanoparticles

- Create strong interfacial interactions with the rubber matrix

- Enhance both mechanical properties and anti-UV aging resistance

In SBR composites reinforced with such hybrid fillers, tensile strength and elongation at break increased by 15.0% and 25.1%, respectively, compared to carbon black-reinforced composites .

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5. Conclusions

The reinforcement of NBR rubber with nano-TiO_2 is governed by a combination of factors that operate synergistically:

1. **The polar nature of NBR** provides favorable interactions with the hydroxylated surface of TiO_2 nanoparticles, enabling effective dispersion and interfacial adhesion.

2. **Surface modification** of TiO_2 nanoparticles through combined covalent and noncovalent approaches can significantly enhance dispersion and interfacial interactions, leading to superior mechanical performance.

3. **Nano-TiO_2 acts as a co-activator** in sulfur curing systems, accelerating crosslinking reactions and increasing crosslink density as evidenced by increased torque differences.

4. **The optimal TiO_2 loading** for maximum mechanical properties is approximately 90 phr, where tensile strength and elongation at break are substantially enhanced compared to unfilled NBR.

5. **Multiple reinforcement mechanisms** operate simultaneously, including hydrodynamic reinforcement, crosslink density enhancement, bound rubber formation, and crack arrest, contributing to the overall improvement in mechanical properties.

6. **Dynamic mechanical properties** are enhanced through higher storage modulus, increased glass transition temperature, and reduced damping, confirming the reinforcing effect.

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